

Sepsis & Septic Shock — Early Recognition & Resuscitation

Recognize organ dysfunction early → cultures + early antibiotics + balanced crystalloid → norepinephrine for persistent hypoperfusion · source control · reassess, don't autopilot the bundle

1 · Recognize — dysfunction, not just infection

Sepsis = life-threatening organ dysfunction from a dysregulated host response to infection (clinically: suspected infection + acute organ dysfunction). **Septic shock** = sepsis with vasopressors needed to keep MAP ≥ 65 and lactate > 2 despite adequate fluids.

Tool	Use
SOFA / ΔSOFA ≥ 2	Defines organ dysfunction (ICU/quantitative)
Lactate	Marker of hypoperfusion + resuscitation target
qSOFA (retired as screen)	Low sensitivity; don't use to rule out — use NEWS-type screens

Organ dysfunction clues: new hypotension, $\uparrow$ lactate, $\uparrow$ creatinine/oliguria, $\uparrow$ bilirubin, altered mentation, thrombocytopenia, new O₂ requirement.

2 · The first hour — culture, antibiotic, fluid

- **Blood cultures $\times 2$ before antibiotics** — but never delay antibiotics to get them.
- **Antibiotics fast:** broad-spectrum, ideally within 1 h for shock/ high probability; the 2026 SSC allows a brief diagnostic window (up to ~ 3 h) when sepsis is *possible but not probable* and there is no shock.
- **Fluids:** ~ 30 mL/kg **balanced crystalloid** (preferred over normal saline) for sepsis-induced hypoperfusion/lactate ≥ 4 or hypotension, then reassess — don't give it blind.
- Draw a **lactate**; recheck if elevated to guide resuscitation.

Balanced crystalloids (LR, Plasma-Lyte) are now favored over normal saline (less hyperchloremic acidosis, possible mortality signal). Source control beats any antibiotic — find and drain it.

3 · Persistent hypoperfusion — pressors

If MAP stays < 65 after/with fluids: start **norepinephrine first-line**. Don't chase fluids endlessly into a non-responder.

Agent	Role
Norepinephrine	First-line vasopressor; titrate to MAP ≥ 65
Vasopressin	Add-on to reduce norepi dose / catecholamine sparing
Epinephrine	Second-line add-on; consider if cardiac dysfunction
Steroids (hydrocortisone)	If shock persists on escalating pressors

Start norepinephrine early — even peripherally through a good IV — rather than over-resuscitating with liters of fluid. Assess fluid responsiveness dynamically (passive leg raise, pulse-pressure variation), not by CVP.

Source & de-escalate

- **Find the source:** exam, imaging, cultures; **control it** (drain abscess, remove infected line, relieve obstruction) as early as feasible.
- **De-escalate** antibiotics on culture data; stop if infection excluded; use procalcitonin trends to support stopping.
- Typical durations are shorter than tradition (often 5–7 days) for many sources once controlled.

Reassess · Do/Avoid

- **DO** reassess perfusion repeatedly (MAP, lactate clearance, urine output, mentation, cap refill)
- **DO** favor balanced crystalloid and start norepinephrine early for non-responders
- **DO** plan source control and antibiotic de-escalation from day 1
- **AVOID** using qSOFA to rule out sepsis (poor sensitivity)
- **AVOID** reflexive fixed fluid boluses in a fluid-unresponsive or already-wet patient
- **AVOID** routine antipyretics, beta-blockers, or probiotics as sepsis therapy (2026 SSC)